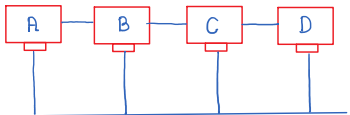


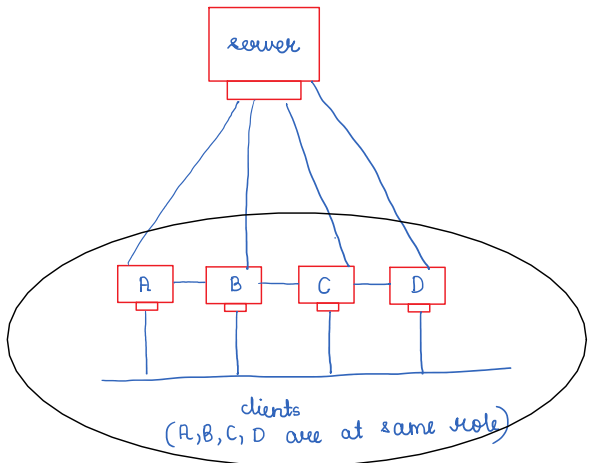
component Role

1. Peer to Peer network:-



A, B, C, D are machines at same role

② Client Server Architecture:-



Ethernet Packet (Frame)

Preamble	SFD	Destination MAC	Source MAC	Type	Data and Pad	FCS
7 Bytes	1 Byte	6 Bytes	6 Bytes	2 Bytes	46-1500 Bytes	4 Bytes
START (1)		C	A	http, https, IPv4, IPv6	M4a	2

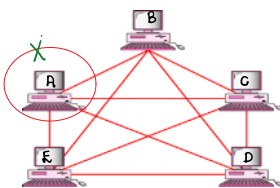
simple to implement
easy to maintain
less cost

cannot be applied for real time situations

priority is not set

Topology:-

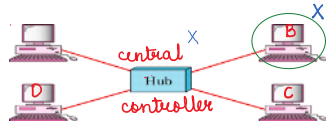
1. Mesh



- point to point
- 5 machines (n)

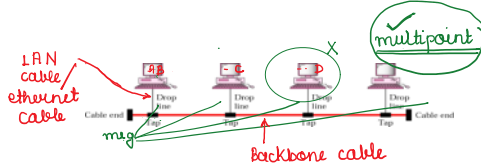
- individual connections = $n-1$
 $= 5-1$
 $= 4 \text{ connect}^{\text{ns}}$
- Applⁿ → military communicatⁿ / aircraft navigation.

2. Star:-



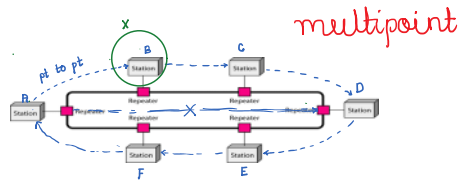
- point to point
- one of the device fails the n/w will not fail
- central controller fails; entire sys fails.

3. Bus



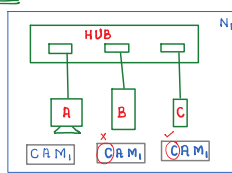
- point to point
- Backbone cable fails only then the entire system will fail.

4. Ring:-



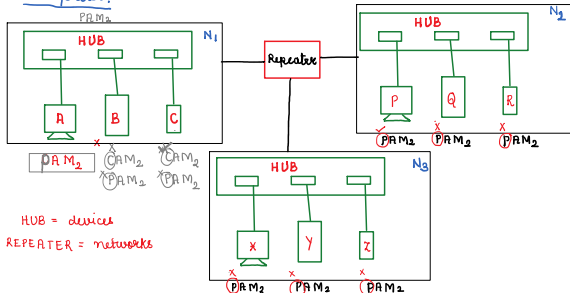
Devices

1. Hub



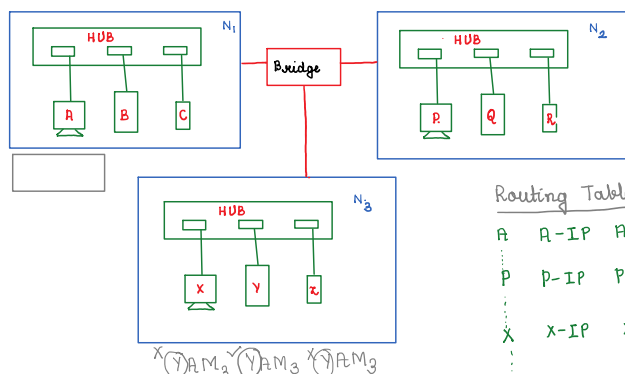
Hub = devices
 ~ simple & less costly

2. Repeater:-



HUB = devices
 REPEATER = networks

3. Bridge



Routing Table:-

A	A-IP	A-MAC	N1
P	P-IP	P-MAC	N2
X	X-IP	X-MAC	N3

Routing Table

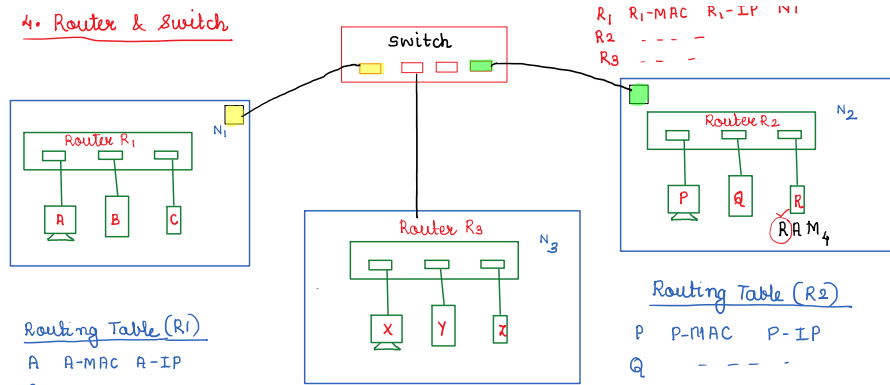
R1	R1-MAC	R1-IP	N1
R2	-	-	-
R3	-	-	-

4. Router & Switch



Router = device

4. Router & Switch



Routing Table (R1)

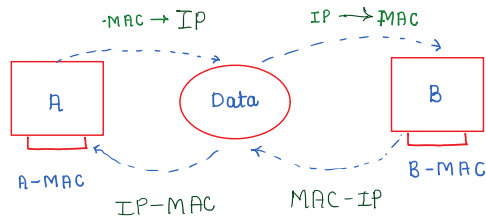
A	A-MAC	A-IP
B	B-MAC	B-IP
C		

Routing Table (R2)

P	P-MAC	P-IP
Q		
R		

ARP (Address Resolution Protocol)

← converts MAC addr to IP addr
IP addr back to MAC addr



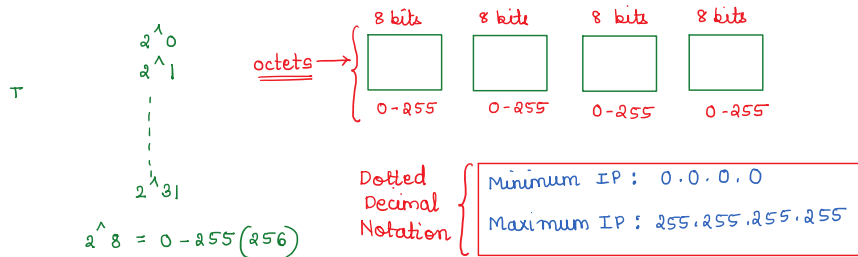
MAC - physical addr
IP - logical addr

IP address:-

IPv4 = 32 bits (4 bytes)

IPv6 = 128 bits (16 bytes)

IPv4 = 32 bits (4 bytes)



Binary notation

Minimum IP: 00000000 00000000 00000000 00000000
Maximum IP: 11111111 11111111 11111111 11111111

